Energy-Limited, Not Control-Limited: An Achievable-Ceiling Analysis of Multi-Hub Vehicle-to-Grid Voltage Support

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***Abstract*—Vehicle-to-grid (V2G) dispatch is widely proposed as a voltage support resource on weak distribution feeders, and a growing literature compares learned controllers against local droop for the task. This paper argues that the comparison is being made before a more basic question has been answered: how much voltage support the connected fleet can supply at all. On a five-hub IEEE 34-bus feeder we compute a per-hour achievable ceiling by optimizing per-hub active and reactive injection directly on the AC power flow, and use it to separate the shortfall owed to control from the shortfall owed to stored energy. Coordinated uneven dispatch across five hubs clears every violating hour at a mild peak, against seven of eighteen hours for a single hub of equal rating. Once realistic availability and state-of-charge limits are applied, however, the fleet delivers only 14.5 percent of the energy the unconstrained controller requests, and the day-integrated violation rises from 4.15 to 42.77 per-unit-hours. Against that, the entire span between a tuned droop law and a soft actor-critic policy trained for 100,000 steps is 0.79 per-unit-hours. Energy adequacy outweighs controller choice by a factor of 45 to 50 at both load levels, which suggests that fleet sizing and availability, not controller sophistication, should govern where effort is spent.**

***Index Terms*—vehicle-to-grid, voltage regulation, distribution networks, reinforcement learning, battery degradation, energy adequacy**

I. Introduction

Distribution feeders serving rural and semi-rural load were never designed for the voltage profiles they are now asked to hold. Long, lightly conductored radial circuits sag under peak demand, and the conventional remedies—reconductoring, additional regulators, capacitor banks—are slow and capital intensive. Against that backdrop, the bidirectional inverters arriving with electric vehicle fleets look like an attractive alternative. They are already being installed, they sit at the load end of the feeder where support is worth most, and they are paid for by someone else. Global electric vehicle stock continues to grow sharply [1], and with it the notional capacity available for grid service.

The research literature has largely accepted this framing and moved on to the control question. A substantial body of work now applies deep reinforcement learning to inverter dispatch for voltage regulation, using safe off-policy methods [3], multi-agent consensus formulations [4], and graph-based architectures with explicit safety layers [5]. Recent work specific to vehicle-to-grid (V2G) extends this to coordinated multi-hub operation with battery-aware constraints [2]. The implicit premise throughout is that the controller is the binding constraint: that a better policy will extract more voltage support from the same hardware.

This paper questions that premise on the specific system where it is most often tested. Our starting point is the observation that none of these studies computes what the fleet could achieve under an ideal controller, so no result—positive or negative—can be placed on a scale. A learned policy that halves the violation of a droop baseline may have captured most of the available benefit or a negligible fraction of it, and the published numbers do not distinguish the two cases.

We therefore build the missing reference. For each hour of a daily load profile we solve for the per-hub active and reactive injection that minimizes a two-sided voltage violation metric, subject only to inverter rating, directly on the AC power flow. This is an energy-unconstrained ceiling: it says what the installed power electronics could do if the batteries behind them were bottomless. Comparing it against droop and learned controllers operating with and without a realistic fleet model then decomposes the total shortfall into a part owed to stored energy and a part owed to control.

The decomposition is lopsided. At a mild peak the ceiling clears every violating hour, an energy-unconstrained droop law captures ninety-one percent of the available improvement, and the same droop law with a realistic fleet captures six percent. The gap between droop and a converged learned policy is smaller than the width of that shortfall by nearly two orders of magnitude. The same pattern holds under aggressive stress.

The contributions are as follows.

- A per-hour achievable ceiling for multi-hub V2G voltage support on the IEEE 34-bus feeder, computed by coordinate descent on the AC power flow, and the energy-versus-control decomposition it enables (Section III).
- A measurement of the active/reactive trade-off on a *stored energy* basis rather than an apparent power basis, which

TABLE I
IDENTIFYING THE REGULATOR SETTING FROM THE NO-V2G BASELINE

Setting	Mild		Aggressive	
	Mean	Min	Mean	Min
Reported [2]	0.963	0.907	0.883	0.807
Regulators off (ours)	0.951	**0.903**	0.860	**0.800**
Regulators active (ours)	1.012	0.983	0.911	0.831

is the correct basis when state of charge is drawn down by apparent power (Section III-C).

- A controller comparison under realistic availability and state-of-charge limits, including a training-budget sweep that separates genuine controller limitations from under-training (Section IV).
- A degradation-aware dispatch frontier, and three negative results—on safety projection, state augmentation, and droop fixed-point convergence—that we believe are useful to report (Sections V and VI).

The remainder of this paper is organized as follows. Section II describes the feeder, the fleet and battery model, the evaluation metrics, and the controllers, and reproduces the reference study's baseline. Section III computes the achievable ceiling and the energy–control decomposition. Section IV reports controller performance under fleet constraints, including a training-budget sweep. Section V presents the degradation-aware frontier, Section VI three negative results, and Section VII concludes.

II. STUDY DESIGN

A. Feeder and Hubs

All experiments use the IEEE 34-bus test feeder [6], a 24.9 kV rural circuit whose length makes it a standard stress case for voltage support studies, solved in OpenDSS [7] through its direct Python interface. The single-hub configuration places one 500 kW/400 kVAr hub at bus 890; the multi-hub configuration distributes five such hubs at buses 890, 844, 832, 830 and 860, matching the reference study [2].

A daily load shape drives hours 06:00 through 23:00, scaled by a peak multiplier. We report two operating points throughout: a *mild* peak ($\lambda_{\max} = 1.5$) and an *aggressive* peak ($\lambda_{\max} = 3.0$), following [2].

B. Reproducing the Reference Baseline

The reference study does not state whether the IEEE 34-bus regulators are active, and the answer changes every number that follows. Its reported no-V2G statistics are the fingerprint that settles it. Those statistics are computed from *hourly mean bus voltages*, so we match that convention here only; everywhere else in this paper we evaluate per phase, for the reason given in Section II-D.

Table I identifies the setting unambiguously. With regulators disabled our minimum voltages fall within 0.004 and 0.007 p.u. of the reported values; with them active the mild minimum is off by 0.076 p.u., far outside any plausible modelling difference. Our feeder mean runs 0.012 to 0.023 p.u. below theirs, which we attribute to load allocation details not specified in the reference, so we identify the configuration from the minimum rather than the mean. All subsequent results use disabled regulators.

C. Fleet and Battery Model

Each hub serves a fleet whose availability varies through the day between forty-five and eighty-five percent, drawn per scenario. Vehicles arrive at seventy percent state of charge and may not be discharged below twenty percent. Critically, state of charge is drawn down by *apparent* power, so reactive support costs the same stored energy as active support. This follows the reference formulation and matters for Section III-C.

The distinction between an unconstrained and a constrained fleet runs through the whole paper. The reference study evaluates multi-hub coordination while "assuming sufficient EV capacity at each hub, allowing the analysis to isolate the benefits of multi-hub coordination from availability constraints" [2]. We report both, because the difference between them turns out to be the main result.

D. Metrics

ANSI C84.1 [11] is a per-phase standard, so we evaluate per-phase rather than on bus-average voltages, which mask single-phase excursions on this unbalanced feeder. Our primary metric is the day-integrated two-sided violation

$$\mathrm{IV} = \sum_{h}\sum_{p}\Big[\max(0, v_{\min} - v_{p,h}) + \max(0, v_{p,h} - v_{\max})\Big]\Delta t \tag{1}$$

in per-unit-hours, with $v_{\min} = 0.95$, $v_{\max} = 1.05$ and $\Delta t = 1$ h. Unlike a count of violation hours, (1) does not saturate when two controllers violate in the same hours by different margins, and being two-sided it charges a controller for overvoltage as well as undervoltage.

Every comparison uses common random numbers: all controllers see an identical availability draw, initial state of charge and load realization, and differences are paired. Confidence intervals are 95 percent over the paired scenarios and training seeds.

E. Controllers

The droop baseline is a Volt-Watt/Volt-VAr characteristic with a ± 0.02 p.u. deadband, consistent with IEEE 1547 [10]. Because droop response and feeder voltage are mutually dependent, we do not evaluate it open-loop; we iterate to the fixed point with damped Jacobi updates and a residual convergence test, a detail that matters more than it appears and which we return to in Section VI.

The learned controller is soft actor-critic [8] as implemented in Stable-Baselines3 [9], with two hidden layers of 256 units, learning rate 3×10^{-4} and batch size 256. Episodes span one day, so intertemporal rationing of a finite battery is representable. The state comprises bus voltages, load multiplier, hour of day, and per-hub state of charge and availability; the

TABLE II
ACHIEVABLE CEILING UNDER OPTIMIZED PER-HUB DISPATCH

Configuration	Clearable hours (of 18)	Day-total IV (p.u.·h)	
		Uniform	Optimized
Single hub, mild	7	11.14	10.79
Single hub, aggressive	2	125.84	123.97
Five hubs, mild	**18**	0.29	**0.00**
Five hubs, aggressive	4	28.33	**17.99**

action sets per-hub active and reactive power as a residual on the droop prior.

III. How Much Support Is Physically Available?

A. The Achievable Ceiling

For each hour we search for the per-hub injection (P_i, Q_i) minimizing (1), subject to $\sqrt{P_i^2 + Q_i^2} \leq S_i^{\text{rated}}$, by coordinate descent over a polar grid with refinement, evaluating a full AC power flow at every candidate. No energy constraint is imposed. Non-converged solutions are masked rather than read, since at deep-sag hours the high-injection end of the search lies past the nose of the power-voltage curve, where OpenDSS leaves stale values in its arrays.

Searching per hub rather than uniformly is essential. With five hubs at equal output, lifting the feeder end to 0.95 p.u. drives the near buses past 1.05 p.u., so hours an uneven dispatch clears comfortably are scored unclearable: the five-hub case then scores *worse* than the single-hub case, which is physically impossible. Per-hub search removed the artifact and moved the mild multi-hub result from five clearable hours to eighteen.

Table II gives the result. A single hub, even at full rating, clears seven of eighteen hours at the mild peak and two at the aggressive peak. Five hubs of the same individual rating, dispatched unevenly, clear all eighteen and four respectively. Fig. 2 shows the hourly picture: the reachable envelope sits on or above the 0.95 p.u. limit for the entire mild day, and falls short of it for eleven hours of the aggressive day no matter how the inverters are dispatched. Those eleven hours are a siting and rating problem, not a control problem, and no policy of any kind can clear them.

The value of coordination is now quantifiable rather than asserted. At equal total inverter rating, distributing capacity across five hubs and dispatching it unevenly reduces day-integrated violation by 100 percent at the mild peak and 36.5 percent at the aggressive peak relative to uniform dispatch.

B. The Energy–Control Decomposition

Fig. 1 places the controllers against this ceiling, and is the central result of the paper. Reading the mild panel: the untreated feeder accumulates 45.61 p.u.·h of violation; the ceiling removes all of it; a droop law with unlimited stored energy removes 41.46 of it, or ninety-one percent; the same droop law with a realistic fleet removes 2.84, or six percent. The fleet delivers 1342 kWh against the 9264 kWh the unconstrained law requests—14.5 percent. At the aggressive peak the ratio is 5.7 percent.

The controller span sits inside that shortfall. Between tuned droop and the best learned policy we trained, the difference is 0.79 p.u.·h at the mild peak and 2.58 at the aggressive peak, against energy shortfalls of 38.62 and 115.69 respectively. Energy adequacy outweighs controller choice by a factor of 49 and 45.

TABLE III
FIVE-HUB FEEDER WITH REALISTIC FLEET CONSTRAINTS

Peak	Controller	IV (p.u.·h)	Throughput (kWh)	Paired wins
Mild	No V2G	45.61	0	—
	Droop	**42.77** $\pm$ 0.32	1342	—
	Learned, 20k steps	45.45 $\pm$ 0.38	5842	0/15
	Learned, 100k steps	43.56 $\pm$ 0.27	5588	2/10
Aggressive	No V2G	171.21	0	—
	Droop	**165.83** $\pm$ 0.60	1582	—
	Learned, 20k steps	169.60 $\pm$ 0.60	3227	0/15
	Learned, 100k steps	168.41 $\pm$ 0.36	3176	0/10

C. Active Versus Reactive Power per Unit of Stored Energy

Because state of charge is drawn down by apparent power, the usual reactive-power-is-free intuition does not apply. Sweeping the injection angle at fixed apparent power and fixed loading, we find that active power delivers between 1.25 and 1.49 times the worst-phase voltage rise of reactive power for the same battery energy withdrawn. The voltage-optimal split is 35 to 40 degrees from the active axis at half rating, migrating toward 17 to 26 degrees at full rating as the resistive voltage drop dominates. For reference, the hub rating ratio of 400 kVAr to 500 kW corresponds to 38.7 degrees.

This has a direct design consequence. On a feeder with this R/X character, a V2G hub sized by apparent power and operated at its rating ratio is systematically over-committing reactive power, and paying full battery energy for the privilege.

IV. What Controllers Actually Deliver

A. Multi-Hub Operation Under Fleet Constraints

Table III reports the five-hub feeder with the fleet model active, over three seeds and five paired scenarios. Droop reaches 42.77 p.u.·h at the mild peak and 165.83 at the aggressive peak. The learned policy trained on the conventional 20,000-step budget reaches 45.45 and 169.60, losing to droop in every paired scenario at both load levels.

Two features of the table deserve comment. First, the learned policy consumes 5842 kWh of battery throughput at the mild peak against droop's 1342, and converts that into a *worse* voltage outcome; at the mild peak its result is statistically indistinguishable from installing no V2G at all. Second, both controllers respect the upper voltage limit exactly, at 1.050 p.u., a point we return to in Section VI.

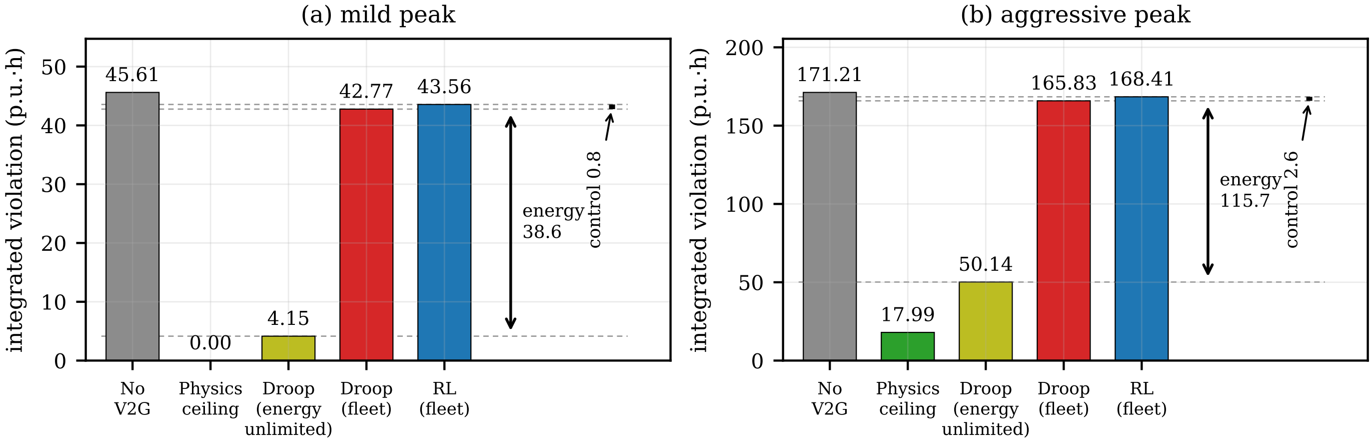

Fig. 1. Day-integrated two-sided voltage violation on the five-hub feeder, decomposed. The physics ceiling is the optimized per-hub dispatch bounded only by inverter rating. The distance between the energy-unlimited and real-fleet droop bars is the cost of fleet energy adequacy; the distance between the two real-fleet bars is the entire span of controller choice.

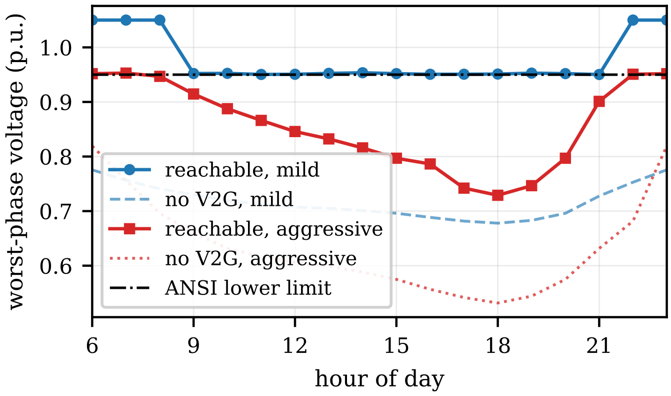

Fig. 2. Worst-phase voltage by hour with no V2G and under optimized per-hub dispatch. The mild-peak envelope clears the limit all day; eleven aggressive-peak hours cannot be cleared at any dispatch.

B. Is the Policy Simply Undertrained?

A negative result about a learned controller invites the obvious objection that it was not trained long enough, and the objection is serious: 20,000 steps is roughly 1100 episodes and 19,000 gradient updates for a ten-dimensional action space. We therefore trained to 100,000 steps and evaluated at 20,000-step checkpoints on identical scenarios, taking care that checkpoints fall on the same chunk boundaries the fixed-budget runs use so that the 20,000-step row measures the same agent the headline table reports.

The answer differs by load level, and Fig. 3 shows why that matters. At the aggressive peak the curve is flat from 40,000 steps onward: violation moves from 169.20 to 168.41 across a fivefold increase in budget, inside the combined confidence interval of 1.13, and droop wins every paired scenario at every checkpoint. Droop's advantage under aggressive stress is not an artifact of training budget.

At the mild peak it is still improving at 100,000 steps. Violation falls from 45.05 to 43.56, a change of -1.48 against a confidence interval of 0.49, the gap to droop narrows monotonically from $+2.27$ to $+0.79$, and the policy takes two of ten paired scenarios—the only scenarios it wins anywhere in this study. We report the mild comparison as unresolved at the budgets tested rather than claiming a droop victory there.

The gap narrowed by 1.48 p.u.·h over a fivefold training increase, against an energy shortfall of 38.62 at the same operating point: more training moves the policy within a band that stays negligible against the constraint governing the outcome.

C. Where the Learned Policy Loses

A scoreboard is less useful than a mechanism, so we examined the active/reactive split the policy actually commands, hour by hour, weighted by apparent power and compared against the voltage-optimal angle recomputed at the injection level the policy itself chose.

The policy does not track the optimum. At the 20,000-step budget the mean absolute deviation is 23.6 degrees at the mild peak and 20.1 degrees at the aggressive peak, against an optimum that sits stably between 30 and 40 degrees for almost the whole day. Its apparent-power-weighted daily mean, 29.2 and 41.5 degrees, sits close to the 38.7-degree hub rating ratio, and the per-hour standard deviation reaches 44.7 degrees. Taken together these are the signature of a controller scaling a roughly fixed power factor rather than allocating between active and reactive power in response to loading. In fifty of 284 sampled hub-hours at the mild peak the policy absorbs reactive power, which is a different action from supplying less of it.

V. Degradation-Aware Dispatch

Battery wear is the standing objection to V2G participation, and the economic case is sensitive to it [12]. The reference formulation carries no battery term in its reward at all. We add an amp-hour throughput cost, expressed on the same scale as

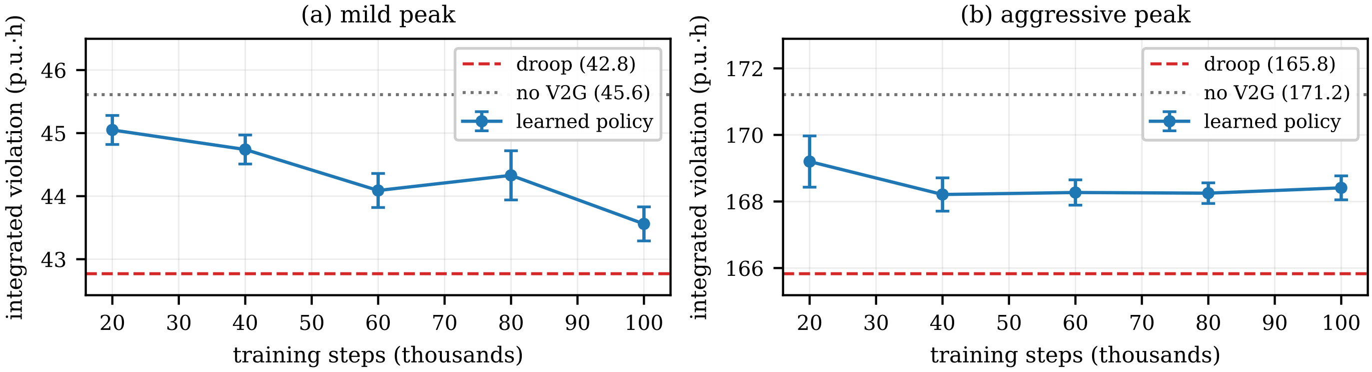

Fig. 3. Violation against training budget, five paired scenarios and two seeds per point. The aggressive-peak curve plateaus by 40,000 steps; the mild-peak curve is still improving at 100,000. Both remain far above the energy-unlimited droop result of 4.15 and 50.14 per-unit-hours respectively.

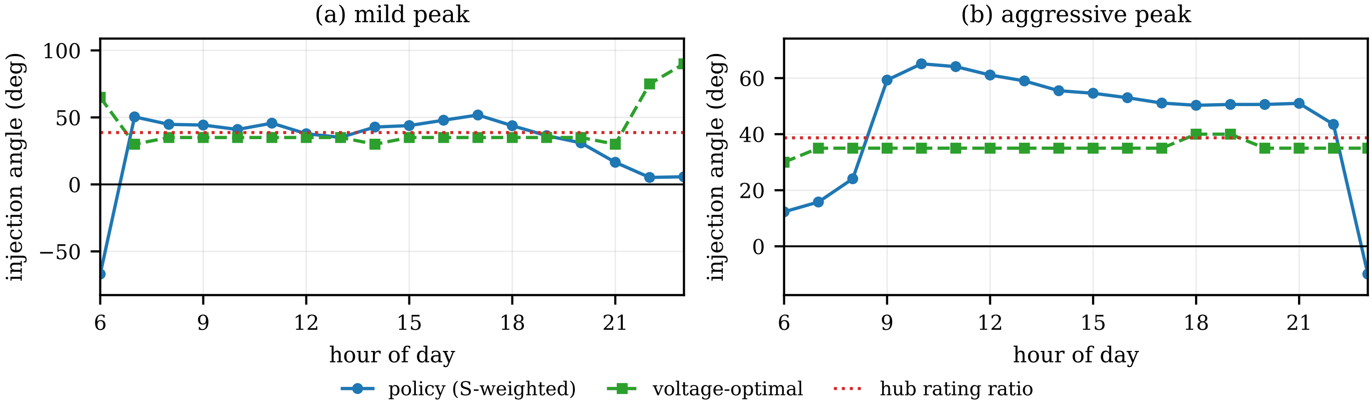

Fig. 4. Commanded injection angle against the voltage-optimal angle, by hour. Zero degrees is pure active power, ninety is pure reactive, and negative values indicate reactive absorption.

the voltage penalty, and sweep its weight across three orders of magnitude.

The frontier is remarkably flat (Fig. 5). Across a 300-fold sweep the day-integrated violation stays within a 2.6 percent band, between 44.83 and 46.01 p.u.·h, while daily battery throughput falls from 6188 to 2449 kWh. Comparing the endpoints, roughly sixty percent of battery throughput can be removed for a 1.1 percent increase in violation.

We read this as a straightforward consequence of the energy-limited regime. When the binding constraint is that the fleet cannot supply what the feeder needs, the marginal voltage value of the last kilowatt-hour discharged is very low, so a wear penalty removes it almost for free. The practical reading is favourable: on feeders in this regime, aggressive degradation-aware dispatch costs almost nothing in voltage performance, which materially improves the participation case for fleet owners.

We caution against over-reading the intermediate points. Each weight was trained with a single seed, and the sequence is not monotone; the defensible claim is the endpoint comparison and the width of the band, not the shape in between.

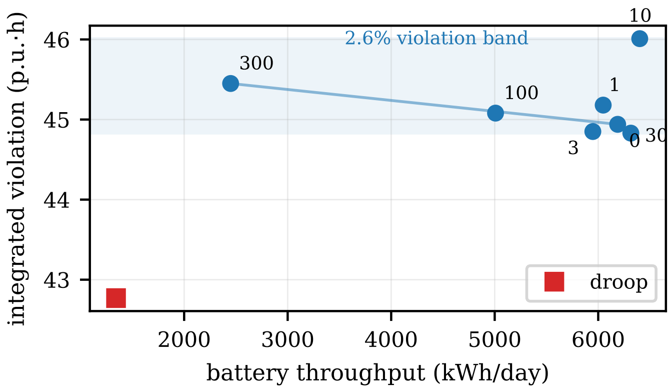

Fig. 5. Violation against battery throughput as the degradation weight is swept from 0 to 300 (labels). Violation stays inside a 2.6 percent band while throughput falls by sixty percent.

VI. Three Things We Checked and Did Not Find

A. A Safety Projection Layer Is Inert Here

Projecting commanded setpoints onto the voltage-feasible set is standard practice in learning-based volt/var control [3], [5], so its absence would be a reasonable reviewer objection. We implemented two projections—a proportional reduction along the commanded ray, and a reactive-first variant motivated by the result of Section III-C—and measured them.

Both are inert. The unprojected policy already reports zero overvoltage and a maximum phase voltage of exactly 1.050 p.u. at both load levels, and projection changes the violation metric by less than the confidence interval. The reason is instructive: the feasibility test acts on commanded setpoints, but the fleet's state-of-charge limit clips those commands first. The energy constraint binds before the voltage constraint does, leaving nothing for a voltage projection to remove. Where overvoltage does appear in our runs, it tracks the training regime rather than the controller: policies trained on independently sampled load multipliers without a fleet in the loop reach 1.229 p.u., while day-structured training with the fleet in the loop holds 1.050 across every configuration and every checkpoint we tested.

B. Fleet State in the Observation Does Not Reduce Violation

Augmenting the observation with state of charge and availability—information the reference formulation does not provide—does not improve voltage performance. At the mild peak the augmented agent reaches 45.41 ± 0.36 against 44.91 ± 0.29 for a voltage-only observation, an overlapping interval. What the augmentation changes is energy: the augmented agent spends 6068 kWh against 2497 kWh for no violation benefit, while the voltage-only agent breaches the upper limit at the aggressive peak where the augmented agent does not.

C. The Droop Fixed Point Does Not Always Converge

Finally, a methodological note that affects any study using a closed-loop droop baseline. Droop response and feeder voltage are mutually dependent, and the fixed point is commonly computed with a fixed iteration count. On the five-hub feeder we found that at a damping factor of 0.6 the iteration does not settle: it enters a period-two limit cycle, with total active power alternating between 1298 and 630 kW at the mild-peak evening hour. A fixed iteration count then returns whichever branch the last iteration happened to land on, and the reported baseline flips with the parity of the iteration count. Damping of 0.3 with a residual test converges in all four configurations we tested. We recommend reporting a convergence diagnostic alongside any droop baseline computed this way.

VII. Conclusion and Future Work

We computed an achievable ceiling for multi-hub V2G voltage support on the IEEE 34-bus feeder and used it to separate what is lost to control from what is lost to stored energy. Uneven coordination across five hubs is genuinely valuable, clearing every violating hour at a mild peak where a single hub of equal rating clears seven of eighteen. But once realistic availability and state-of-charge limits are applied, the fleet supplies well under a fifth of the energy the controller asks for, and the resulting shortfall exceeds the entire span between a tuned droop law and a converged learned policy by a factor of roughly forty-five to fifty.

We do not read this as an argument against learned control in distribution systems. We read it as an argument that the regime should be established before the controller is designed, because in the energy-limited regime a better policy has almost nothing left to win. Both the achievable ceiling and an energy adequacy ratio are computable from feeder and fleet data before any controller exists, and we suggest they be reported as standard context alongside controller comparisons.

Two results point the same way. A degradation-aware objective removes sixty percent of battery throughput for a 1.1 percent violation penalty, precisely because the marginal kilowatt-hour is nearly worthless in this regime—good news for fleet owners weighing participation. And a safety projection layer, standard elsewhere, is measurably inert here because the energy constraint binds first.

The study covers one feeder and one learning algorithm, and the mild-peak controller comparison remains unresolved at the training budgets we could afford. Two extensions follow directly. The ratio we report depends on feeder impedance and hub siting, so it should be recomputed on a larger feeder. More usefully, the ceiling computation is cheap enough to invert: instead of scoring a controller against it, it can serve as a planning screen that returns the fleet size and availability needed to make a target violation level reachable.

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